Medvale Internal Medicine

Chapter 5D

Renal/GU Review and Integrated Cases

Fluids, electrolytes, acid-base, AKI, CKD, glomerular disease, tubular/interstitial disease, stones, obstruction, and high-yield synthesis

Original Medvale synthesis chapter. Uploaded renal source pages and Anki material were used as coverage signals only; explanations, tables, algorithms, cases, and extraction maps are original.

Section role	This is the capstone for Part V: Nephrology, Fluids, and Acid-Base.
Core task	Rapidly classify renal presentations by danger, compartment, nephron segment, urine sediment, time course, and systemic context.
Clinical emphasis	Do not memorize isolated facts. Build a problem representation that links sodium and water, potassium and acid-base, urine sediment, GFR trajectory, and systemic clues.
Graph-RAG emphasis	Each disease is mapped to nodes, edges, evidence triggers, laboratory anchors, and management links.

Chapter map

Unit	What it answers
1. Integrated renal mental model	How to decide whether a case is primarily volume/osmotic, electrolyte, acid-base, filtration, sediment, systemic immune, tubular, or obstructive.
2. Emergency triage	Which renal problems kill in minutes to hours and what action comes first.
3. Diagnostic algorithms	Sodium/water, potassium, acid-base, AKI, urinalysis, glomerular disease, stones/obstruction.
4. High-yield comparison tables	FeNa/FeUrea, urine sediment, nephritic vs nephrotic, complement patterns, RTA patterns, dialysis indications.
5. Integrated cases	Cases that combine multiple renal mechanisms rather than testing one chapter at a time.
6. Graph-RAG extraction map	Node classes, relationships, facts, and retrieval anchors for future knowledge graph ingestion.

Clinical source context

The uploaded renal disease source emphasizes AKI categories, urine sediment, FeNa/FeUrea, prerenal versus intrinsic versus postrenal logic, CKD complications, and dialysis indications. The earlier volume/electrolyte/acid-base source emphasizes total body water compartments, sodium/water disorders, correction rates, potassium emergencies, calcium/phosphate/magnesium physiology, and acid-base interpretation. This review deliberately connects those pieces into mixed clinical presentations.

1. Integrated renal mental model

Renal questions become easier when every vignette is forced through a small set of physiologic gates. First ask whether the patient is unstable: arrhythmia from potassium, seizure or coma from sodium or uremia, pulmonary edema from volume overload, severe acidemia, malignant hypertension, sepsis, obstruction in a solitary kidney, or rapidly progressive glomerulonephritis. Then determine whether the dominant problem is water, salt, solute clearance, urine sediment, nephron segment function, or urinary drainage. The same patient may have several gates open at once: for example, septic shock can produce prerenal azotemia, ATN, lactic acidosis, hyperkalemia, and respiratory alkalosis before creatinine has fully risen.

A useful renal illness script has five linked elements: time course, volume status, serum chemistry, urine behavior, and systemic context. Time course separates acute kidney injury from chronic kidney disease and acute-on-chronic disease. Volume status explains ADH, RAAS, urine sodium, edema, and prerenal physiology. Serum chemistry reveals dangerous electrolytes and acid-base patterns. Urine behavior, especially osmolality, sodium, protein, blood, and casts, points to a nephron segment or a glomerular inflammatory pattern. Systemic context - diabetes, hypertension, lupus, infection, malignancy, drug exposure, obstruction, stones, or pregnancy - explains why the kidney is injured.

Renal axis	Question	High-yield anchors	Common trap
Water/tonicity	Is sodium abnormal because water is too high or too low?	Serum osmolality, glucose correction, urine osmolality, urine Na, symptoms, correction rate.	Treating the sodium number before identifying hypotonic, isotonic, or hypertonic hyponatremia.
Effective arterial blood volume	Does the kidney perceive underfilling?	Orthostasis, JVP, edema, cirrhosis/heart failure, urine Na, FeNa/FeUrea, BUN/Cr.	Confusing total-body edema with effective perfusion. CHF and cirrhosis can be total-body overloaded but intravascularly underfilled.
Filtration	Is GFR acutely or chronically reduced?	Creatinine trend, urine output, CKD duration >3 months, kidney size/cortical thinning, albuminuria.	Assuming a single creatinine defines chronicity.
Sediment	Is the urine bland, muddy, inflammatory, or glomerular?	Hyaline casts, granular casts, RBC casts/dysmorphic RBCs, WBC casts, eosinophiluria, crystals.	Overinterpreting FeNa when diuretics, CKD, contrast, or mixed disease are present.
Segment	Which nephron segment is failing?	Glomerulus: protein/blood/casts; tubule: ATN/RTA/electrolytes; interstitium: fever/rash/eosinophils/WBCs; collecting duct: ADH/aldosterone effects.	Calling every creatinine rise ATN without examining urine and obstruction.
Drainage	Can urine leave the kidney?	Bladder scan, Foley response, hydronephrosis, BPH, stones, pelvic mass, solitary kidney.	Missing postrenal AKI because the patient is still making some urine.

The one-minute renal presentation

A disciplined presentation can sound like this: "This is an acute-on-chronic kidney injury with oliguria, hyperkalemic non-anion-gap metabolic acidosis, muddy brown casts, and a recent hypotensive insult, most consistent with ischemic ATN superimposed on diabetic CKD. There is no hydronephrosis, but there is pulmonary edema, so I am stopping nephrotoxins, treating potassium, restricting potassium/phosphate, dosing medications for GFR, and assessing need for dialysis using AEIOU." That sentence links syndrome, mechanism, danger, and action.

2. Emergency triage: what to act on before finishing the differential

Renal review cases often hide immediate management inside diagnostic reasoning. Some abnormalities can be investigated after stabilization; others require treatment while the workup is being collected. Severe hyperkalemia with ECG changes is treated before the cause is known. Severe symptomatic hyponatremia is treated with hypertonic saline while limiting the rise in serum sodium. Pulmonary edema with hypoxemia in kidney failure may require noninvasive ventilation, diuretics if residual function exists, and dialysis if refractory. Obstruction with infection or solitary kidney is not a routine outpatient stone; it is urgent decompression.

Danger pattern	Clues	Immediate move	Mechanism
Hyperkalemic membrane instability	Peaked T waves, PR prolongation, QRS widening, sine wave, weakness, K often >6.0 but ECG matters more than number.	IV calcium for membrane stabilization, then insulin/dextrose, beta agonist, bicarbonate if acidemic, remove K with diuresis/resin/dialysis.	Reduced K excretion, acidosis shifting K out of cells, tissue breakdown, RAAS blockade, hypoaldosteronism.
Severe symptomatic hyponatremia	Seizure, coma, severe confusion, vomiting, headache, Na often <120 but symptoms and acuity matter.	Hypertonic saline bolus strategy; frequent Na checks; avoid overcorrection, especially alcoholism, malnutrition, liver disease, hypokalemia.	Low tonicity drives water into brain cells; rapid correction risks osmotic demyelination.
Severe hypernatremia or rapid rise	Altered mental status, weakness, seizures; often water loss in DI, osmotic diuresis, poor access to water.	If hypotensive, restore perfusion with isotonic fluid first; then replace free water gradually.	Brain cell dehydration; overly fast correction can cause cerebral edema.
Severe acidemia	pH near or below 7.1, shock, hyperkalemia, impaired contractility, toxic alcohol, renal failure.	Treat cause; consider bicarbonate selectively; dialysis for refractory acidosis or toxins.	Hydrogen ion excess impairs cardiac and enzymatic function; compensation may fail.
Pulmonary edema in kidney failure	Hypoxemia, crackles, high JVP, severe hypertension, oliguria/anuria.	Oxygen/NIPPV, loop diuretic if responsive, vasodilators if hypertensive, dialysis if refractory.	Inability to excrete sodium/water plus high afterload or low cardiac reserve.
Obstructed infected system	Fever, flank pain, pyuria, hydronephrosis, stone, sepsis, solitary kidney.	Antibiotics plus urgent urologic drainage.	Pressure and infection behind obstruction can rapidly destroy renal function and cause sepsis.
Rapidly progressive GN	RBC casts, dysmorphic RBCs, hypertension, rising creatinine over days-weeks, pulmonary hemorrhage/purpura/sinus disease.	Urgent nephrology; serologies and biopsy; immunosuppression often cannot wait if severe.	Inflammatory glomerular capillary injury with crescent formation.

Memory anchor: renal emergencies are often electrical, osmotic, pressure, acid, toxin, or immune emergencies. Electrical = potassium; osmotic = sodium/water; pressure = pulmonary edema or obstruction; acid = severe metabolic acidosis; toxin = dialyzable poisons; immune = rapidly progressive GN.

3. Diagnostic algorithms

Algorithm A: sodium and water disorders

Step	Decision	Interpretation/action
1	Measure serum osmolality and check glucose, lipids, protein if needed.	Low osmolality = true hypotonic hyponatremia. Normal/high osmolality suggests pseudohyponatremia or osmotic shift from glucose/mannitol.
2	If hypotonic hyponatremia, assess symptom severity and chronicity.	Seizure/coma/severe symptoms require hypertonic saline with close monitoring. Chronic mild cases need cause-specific correction and fluid strategy.
3	Use urine osmolality.	Uosm <100 mOsm/kg suggests suppressed ADH: primary polydipsia or low-solute intake. Uosm >100 suggests ADH effect.
4	Use urine sodium and volume status.	Low urine Na often means low effective arterial volume; high urine Na suggests renal salt loss, diuretics, adrenal insufficiency, SIADH, or kidney disease.
5	For hypernatremia, decide whether hypovolemic, euvolemic, or hypervolemic.	Hypovolemic = salt and water loss with more water loss. Euvolemic = water loss, often DI or respiratory/skin losses. Hypervolemic = sodium gain.
6	Correct safely.	Restore perfusion first when unstable. Then correct tonicity slowly enough to prevent cerebral edema or osmotic demyelination.

Algorithm B: acid-base disorders

Step	Decision	Key formula or clue
1	Confirm pH direction.	pH <7.35 acidemia; pH >7.45 alkalemia. Mixed disorders can have near-normal pH.
2	Identify primary process from HCO3- and PaCO2.	Low HCO3- = metabolic acidosis; high HCO3- = metabolic alkalosis; high PaCO2 = respiratory acidosis; low PaCO2 = respiratory alkalosis.
3	Check compensation.	Winter formula for metabolic acidosis: expected PaCO2 = 1.5 x HCO3- + 8 +/- 2. Metabolic alkalosis expected PaCO2 ~ 0.7 x (HCO3- - 24) + 40 +/- 5. Acute respiratory acidosis: HCO3- rises ~1 per 10 PaCO2; chronic rises ~3.5-4. Acute respiratory alkalosis: HCO3- falls ~2 per 10; chronic falls ~4-5.
4	If metabolic acidosis, calculate anion gap.	AG = Na - (Cl + HCO3). Correct upward for low albumin: add ~2.5 mEq/L to AG for each 1 g/dL albumin below 4.
5	If high AG, check delta ratio and osmolar gap.	Delta ratio = (AG - 12)/(24 - HCO3). Low ratio suggests concurrent non-gap acidosis; high ratio suggests concurrent metabolic alkalosis or chronic respiratory acidosis. Osmolar gap supports toxic alcohols or unmeasured osmoles.
6	Treat mechanism, not just pH.	Shock/lactate, ketoacidosis, renal failure, RTA, diarrhea, vomiting/diuretics, hypoventilation, salicylates, sepsis, pregnancy, liver disease.

Algorithm C: AKI and urinalysis

Step	Question	Most useful clues
1	Is this AKI, CKD, or acute-on-chronic?	Creatinine baseline and trend, eGFR duration, prior albuminuria, anemia/mineral bone disease, kidney size/cortical thinning, history of diabetes/HTN.
2	Is urine output low?	Oliguria <0.5 mL/kg/hr is clinically important; anuria suggests obstruction, severe shock, renal vascular catastrophe, or cortical necrosis.
3	Could it be postrenal?	Bladder scan and Foley; renal ultrasound for hydronephrosis; suspect BPH, stones, pelvic mass, neurogenic bladder, retroperitoneal fibrosis.
4	Could it be prerenal?	Low effective arterial volume, high BUN/Cr, concentrated urine, low urine Na/FeNa, response to perfusion correction.
5	Is there intrinsic renal injury?	Muddy brown casts = ATN; RBC casts/dysmorphic RBCs = GN; WBC casts = pyelonephritis/AIN; heavy protein = glomerular; crystals = stone/drug/tumor lysis.

Step	Question	Most useful clues
6	What must be stopped or adjusted?	NSAIDs, ACEi/ARB during unstable perfusion, diuretics if volume depleted, contrast, aminoglycosides, vancomycin, amphotericin, chemotherapy; dose all renally cleared drugs.
7	Does the patient need urgent dialysis?	AEIOU: Acidosis, Electrolytes, Intoxications, Overload, Uremia.

4. High-yield integrated tables

Volume, sodium, and urine studies

Volume status is never a single physical exam sign. It is a synthesis of history, intake/output, daily weights, blood pressure/orthostasis, mucous membranes, edema/JVP, lung exam, kidney perfusion markers, and urine studies. Serum sodium reflects water balance more than total body sodium, while edema reflects sodium retention and hydrostatic/oncotic forces. The exam can be misleading: cirrhosis, nephrotic syndrome, and heart failure may create edema while the kidney behaves as if arterial volume is low.

Pattern	Serum/urine pattern	Mechanism	Typical examples
Hypovolemic hyponatremia	Low Na; Uosm usually >100; urine Na low if extrarenal loss, higher with renal salt loss/diuretics.	ADH is appropriately high because effective arterial volume is low; water is retained out of proportion to salt.	Vomiting, diarrhea, burns, third spacing, diuretics, mineralocorticoid deficiency.
Euvolemic hyponatremia	Low Na; Uosm >100 in SIADH; urine Na often >30 if intake adequate.	ADH effect without obvious edema or depletion; solute intake may be limiting in beer potomania/tea-toast.	SIADH, hypothyroidism, adrenal insufficiency, postop pain/nausea, drugs, pulmonary/CNS disease.
Hypervolemic hyponatremia	Low Na with edema; urine Na often low unless renal failure/diuretics.	Low effective arterial volume triggers ADH and RAAS despite total-body fluid overload.	Heart failure, cirrhosis, nephrotic syndrome, advanced kidney failure.
Hypovolemic hypernatremia	High Na, high serum osmolality, signs of volume depletion.	Loss of water exceeds loss of sodium.	Osmotic diuresis, diarrhea, sweating, burns, loop diuretics.
Euvolemic hypernatremia	High Na with less obvious ECF contraction.	Pure water loss or inability to concentrate urine.	Central DI, nephrogenic DI, insensible losses, impaired thirst/access to water.
Hypervolemic hypernatremia	High Na with edema or iatrogenic sodium load.	Sodium gain exceeds water gain.	Hypertonic saline, sodium bicarbonate, mineralocorticoid excess, saltwater ingestion.

Potassium and acid-base integration

Finding	Think about	First clinical question
Hyperkalemia + metabolic acidosis	AKI/CKD, hypoaldosteronism/type 4 RTA, DKA, tissue breakdown, ACEi/ARB/MRA, TMP-SMX.	Is there ECG instability requiring IV calcium now?
Hypokalemia + metabolic alkalosis + hypertension	Mineralocorticoid excess, primary aldosteronism, renovascular disease, Cushing, licorice.	Is urine chloride high and blood pressure elevated?
Hypokalemia + metabolic alkalosis + low/normal BP	Vomiting, NG suction, diuretics, Bartter/Gitelman syndromes.	Is urine chloride low (vomiting) or high (diuretic/tubular)?
Hypokalemia + non-anion-gap acidosis	Diarrhea, proximal RTA, distal RTA.	Is urine anion gap positive and urine pH inappropriately high?
Hyperkalemia + non-anion-gap acidosis	Type 4 RTA, diabetic hyporeninemic hypoaldosteronism, adrenal insufficiency, RAAS blockade.	Is aldosterone effect low or blocked?
Low Mg + refractory low K or low Ca	Alcohol use, diarrhea, PPIs, aminoglycosides, cisplatin, poor intake.	Replace Mg or K/Ca may not correct.

Compensation formulas quick table

Primary disorder	Expected compensation	What abnormal compensation means
Metabolic acidosis	PaCO2 = 1.5 x HCO3- + 8 +/- 2	Higher PaCO2 = concurrent respiratory acidosis; lower PaCO2 = concurrent respiratory alkalosis.
Metabolic alkalosis	PaCO2 ~ 0.7 x (HCO3- - 24) + 40 +/- 5	Lower PaCO2 = respiratory alkalosis; much higher PaCO2 = respiratory acidosis.
Acute respiratory acidosis	HCO3- rises ~1 mEq/L per 10 mm Hg PaCO2 above 40	More HCO3- suggests chronicity or metabolic alkalosis.

Primary disorder	Expected compensation	What abnormal compensation means
Chronic respiratory acidosis	HCO_3^- rises ~3.5-4 mEq/L per 10 mm Hg $PaCO_2$ above 40	Less HCO_3^- suggests acute component or metabolic acidosis.
Acute respiratory alkalosis	HCO_3^- falls ~2 mEq/L per 10 mm Hg $PaCO_2$ below 40	Lower HCO_3^- suggests metabolic acidosis.
Chronic respiratory alkalosis	HCO_3^- falls ~4-5 mEq/L per 10 mm Hg $PaCO_2$ below 40	Higher HCO_3^- suggests metabolic alkalosis.

AKI: prerenal, intrinsic, postrenal

AKI classification is not merely academic because the first treatment differs: restore perfusion, stop injury and support recovery, or relieve obstruction. The categories also overlap. Prolonged prerenal physiology can become ischemic ATN; postrenal obstruction can become tubular injury; glomerulonephritis can present as intrinsic AKI with hypertension and edema.

Feature	Prerenal azotemia	Acute tubular necrosis	Glomerulonephritis	AIN/pyelonephritis	Postrenal obstruction
Primary problem	Low renal perfusion with intact tubules.	Tubular epithelial injury from ischemia/toxins.	Inflamed glomerular capillaries.	Interstitial inflammation or kidney infection.	Urine outflow pressure blocks filtration.
Urine sediment	Bland, hyaline casts.	Muddy brown granular casts; renal tubular epithelial cells.	RBC casts, dysmorphic RBCs, protein.	WBC casts; pyuria; eosinophils sometimes in AIN.	Bland or hematuria/pyuria depending on cause.
Urine Na/FeNa	Low urine Na; FeNa often <1%.	Urine Na high; FeNa often >2%.	Variable.	Variable.	Variable; early may mimic prerenal.
Urine osmolality	Concentrated, often >500 mOsm/kg.	Poorly concentrated, often <350 mOsm/kg.	Variable.	Variable.	Variable.
Best first tests	Volume/perfusion assessment, med review.	Sediment, exposure review, CK/contrast/toxins.	UA microscopy, protein quantification, complements/serologies, biopsy when indicated.	UA/culture, drug review, eosinophilia, imaging if infection/obstruction concern.	Bladder scan, Foley, renal ultrasound.
First management move	Restore effective perfusion; hold nephrotoxins.	Supportive care; remove insult; manage complications.	Urgent nephrology; disease-specific immunologic workup and therapy.	Stop culprit drug; antibiotics for infection; steroids in select AIN.	Drain bladder/upper tract; treat infection urgently.

Urinalysis pattern recognition

Urine clue	Meaning	Diseases to connect
Hyaline casts	Concentrated urine and low flow; nonspecific but fits prerenal physiology.	Dehydration, heart failure, cirrhosis, diuretics, exercise.
Muddy brown granular casts	Tubular epithelial injury.	Ischemic ATN, nephrotoxic ATN, pigment nephropathy.
RBC casts/dysmorphic RBCs	Glomerular bleeding.	IgA nephropathy, postinfectious GN, lupus nephritis, ANCA vasculitis, anti-GBM, MPGN/C3.
WBC casts	Inflammation in kidney parenchyma.	Pyelonephritis, AIN, sometimes glomerulonephritis.
Oval fat bodies/fatty casts	Lipiduria from heavy protein leak.	Nephrotic syndrome: MCD, FSGS, membranous, diabetic kidney disease, amyloid.
Crystals	Supersaturation, drug precipitation, tumor lysis, ethylene glycol.	Calcium oxalate/phosphate, uric acid, struvite, cystine, acyclovir/indinavir, ethylene glycol.
Heavy albuminuria	Glomerular permeability problem.	Diabetic kidney disease, membranous nephropathy, FSGS, MCD, lupus.
Sterile pyuria	Inflammation without routine culture growth.	AIN, stones, TB, STIs, partially treated UTI, interstitial cystitis.

CKD: chronicity, complications, and slowing progression

CKD is a chronic risk state, not just a creatinine label. The modern frame is cause, GFR category, and albuminuria category. Diabetes and hypertension remain common causes, but a CKD review should also search for glomerular disease, obstruction, cystic disease, recurrent infection, medications, and systemic inflammatory disease. Complications emerge because the kidney loses endocrine, excretory, acid-base, electrolyte, and volume functions.

CKD complication	Mechanism	Clinical/lab pattern	Management link
Hypertension/volume overload	Sodium retention, RAAS activation, reduced natriuresis.	Edema, pulmonary congestion, LVH, resistant HTN.	Sodium restriction, diuretics, RAAS blockade when appropriate, dialysis if refractory.

CKD complication	Mechanism	Clinical/lab pattern	Management link
Hyperkalemia	Reduced distal K secretion; RAAS blockade; type 4 RTA.	Weakness, ECG changes, arrhythmia risk.	Diet/med review, diuretics, binders, bicarbonate if acidotic, dialysis for severe/refractory.
Metabolic acidosis	Reduced ammoniagenesis and acid excretion.	Low HCO_3^-, bone/muscle effects, hyperkalemia.	Alkali therapy when indicated; treat kidney failure progression.
Mineral bone disease	Phosphate retention, low calcitriol, secondary hyperparathyroidism.	High phosphate, low/normal Ca, high PTH, bone pain/fracture risk, vascular calcification.	Phosphate restriction/binders, vitamin D strategies, calcimimetics in selected patients.
Anemia	Reduced erythropoietin plus iron/inflammation.	Normocytic anemia, fatigue, dyspnea.	Iron assessment/repletion, ESA in selected patients, avoid transfusion when transplant candidate if possible.
Uremic symptoms	Accumulation of toxins and inflammation.	Anorexia, nausea, pruritus, pericarditis, encephalopathy, platelet dysfunction.	Dialysis when symptomatic or with emergency indications.
Medication toxicity	Reduced clearance and altered protein binding.	Unexpected sedation, bleeding, hypoglycemia, antibiotic toxicity.	Renal dosing, avoid nephrotoxins, medication reconciliation.

Glomerular disease synthesis

Glomerular disorders are best organized by syndrome and immune pattern. Nephritic disease is an inflammatory capillary problem: hematuria, RBC casts, hypertension, edema, reduced GFR, and usually subnephrotic to moderate proteinuria. Nephrotic disease is a filtration barrier problem: heavy proteinuria, hypoalbuminemia, edema, hyperlipidemia, lipiduria, and thrombotic/infectious risk. Many diseases can mix features, especially lupus nephritis, membranoproliferative patterns, advanced diabetic kidney disease, and rapidly progressive GN.

Disease/pattern	Syndrome	Key clue	Complement pattern	Management logic
Minimal change disease	Nephrotic	Children; abrupt edema; selective albuminuria; NSAIDs/Hodgkin association in adults.	Usually normal.	Steroid responsive; biopsy more common in adults.
FSGS	Nephrotic or mixed	Obesity, HIV, heroin, reduced nephron mass, APOL1 risk, steroid resistance.	Usually normal.	Treat secondary cause; RAAS/SGLT2 supportive; immunosuppression when primary.
Membranous nephropathy	Nephrotic	Adult nephrotic syndrome; PLA2R; malignancy, HBV, lupus, drugs.	Usually normal unless secondary lupus.	Risk-stratify; supportive therapy; immunosuppression for high risk.
Diabetic kidney disease	Albuminuric CKD, often nephrotic-range later	Long diabetes, retinopathy, progressive albuminuria then GFR loss.	Normal.	Glycemic/BP control, RAAS blockade, SGLT2 inhibitor when eligible, finerenone in selected cases.
IgA nephropathy	Nephritic +/- proteinuria	Synpharyngitic hematuria within days of URI.	Usually normal.	Risk stratify by proteinuria/GFR; supportive therapy first; selected immunosuppression.
Postinfectious GN	Nephritic	Hematuria/edema after infection; low C3; subepithelial humps.	Low C3, often returns to normal.	Treat infection/supportive; monitor recovery.
Lupus nephritis	Nephritic, nephrotic, or mixed	SLE features; anti-dsDNA; active urine sediment.	Low C3 and C4 common.	Biopsy class guides immunosuppression.
ANCA vasculitis	Rapidly progressive nephritic	Sinus/lung/skin/neuropathy; pauci-immune crescents.	Usually normal.	Urgent immunosuppression; consider plasma exchange in select severe cases.
Anti-GBM disease	RPGN +/- pulmonary hemorrhage	Hemoptysis + nephritic AKI; linear IgG.	Usually normal.	Plasma exchange plus immunosuppression if salvageable.
MPGN/C3 glomerulopathy	Mixed nephritic-nephrotic	Low complement, chronic infections, monoclonal gammopathy, complement dysregulation.	Often low C3; C4 variable.	Treat cause; biopsy and complement/hematologic workup.
Cryoglobulinemic GN	Mixed nephritic-nephrotic	Purpura, neuropathy, arthralgia, hepatitis C, low complement.	Low C4 prominent.	Treat HCV/underlying clone; immunosuppression in severe disease.

Tubular, interstitial, stone, and obstructive synthesis

Problem	Clinical script	Diagnostic anchor	Management anchor
ATN	AKI after shock, sepsis, pigment, contrast, aminoglycoside, amphotericin, cisplatin, or severe prerenal state.	Muddy brown casts, FeNa often high, poor concentrating ability, delayed recovery.	Supportive care, remove insult, manage electrolytes/volume, dialysis if AEIOU.
AIN	AKI days-weeks after medication or infection; fever/rash/eosinophilia may be absent.	Pyuria/WBC casts, sterile pyuria, eosinophiluria imperfect, biopsy if uncertain.	Stop culprit; consider steroids if significant and persistent after infection excluded.
Papillary necrosis	Flank pain/hematuria/obstruction with sloughed papillae.	Sickle cell disease/trait, analgesic nephropathy, diabetes, pyelonephritis.	Treat cause, manage obstruction/infection.
RTA	Non-anion-gap metabolic acidosis out of proportion to GFR.	Type 1 distal: urine pH high, stones, low K. Type 2 proximal: bicarbonaturia, low K. Type 4: hyperK, hypoaldosterone.	Alkali and electrolyte strategy; treat autoimmune/drug/diabetic/adrenal cause.

Problem	Clinical script	Diagnostic anchor	Management anchor
Nephrolithiasis	Colicky flank pain radiating to groin, hematuria, nausea/vomiting.	Noncontrast CT most sensitive; ultrasound in pregnancy/children or radiation avoidance.	Analgesia, hydration to thirst, alpha blocker in selected ureteral stones, urgent drainage if infected/obstructed solitary kidney/AKI/refractory pain.
Pyelonephritis	Fever, flank pain, pyuria, systemic symptoms; may cause WBC casts.	UA/culture; imaging if sepsis, obstruction, stones, poor response, recurrent disease.	Antibiotics; source control if obstruction/abscess.
Obstructive uropathy	AKI, hydronephrosis, LUTS, stone, pelvic mass, neurogenic bladder; may be painless.	Bladder scan/Foley; renal ultrasound; CT if stone/mass.	Decompression; monitor post-obstructive diuresis and electrolytes.
ADPKD	Family history, enlarged kidneys, HTN, hematuria, stones, cyst infection, cerebral aneurysm association.	Ultrasound/CT/MRI cyst burden by age and family history.	BP control, avoid nephrotoxins, treat complications, consider tolvaptan in selected rapid progressors.

Dialysis indications: AEIOU with renal nuance

Letter	Indication	Examples
A - Acidosis	Severe metabolic acidosis refractory to medical therapy.	Renal failure with pH severe enough to impair hemodynamics; toxic alcohol acidosis.
E - Electrolytes	Severe or refractory electrolyte disturbance.	Hyperkalemia with ECG changes or persistent elevation despite temporizing/removal attempts.
I - Intoxications	Dialyzable toxins.	Ethylene glycol, methanol, lithium, salicylates in selected severe cases.
O - Overload	Volume overload refractory to diuretics or causing respiratory failure.	Pulmonary edema with hypoxemia in oliguric/anuric kidney failure.
U - Uremia	Symptomatic toxin accumulation.	Pericarditis, encephalopathy, seizures, severe bleeding/platelet dysfunction, intractable nausea/vomiting.

5. Integrated cases

These cases are intentionally mixed. For each, identify the immediate danger, the renal syndrome, the mechanism, and the next best action.

Case 1 - Oliguric sepsis with rising creatinine

Stem: A 68-year-old with pneumonia becomes hypotensive overnight. Creatinine rises from 0.9 to 2.1 mg/dL, urine output is 0.3 mL/kg/hr, lactate is elevated, potassium is 5.8, and urine microscopy later shows muddy brown granular casts.

Teaching explanation: Immediate danger: shock, hyperkalemia trend, acidosis risk. Syndrome: early prerenal physiology progressing to ischemic ATN. Next action: restore perfusion with appropriate resuscitation/vasopressors, stop nephrotoxins, monitor K/acid-base/urine output, and prepare for dialysis only if AEIOU emerges. Muddy casts support intrinsic tubular injury after the hemodynamic insult.

Case 2 - Edematous hyponatremia in heart failure

Stem: A patient with severe systolic heart failure has Na 124, serum osmolality low, Uosm 520, urine Na 12, JVP elevation, edema, and dyspnea.

Teaching explanation: The kidney perceives low effective arterial blood volume despite total-body fluid excess. ADH and RAAS retain water and sodium, with water retention dominating the sodium concentration. Treat the heart failure/volume overload; fluid restriction and loop diuresis are typical, with urgent hypertonic saline reserved for severe neurologic symptoms.

Case 3 - Vomiting, alkalosis, and kidney injury

Stem: A patient has several days of vomiting, orthostasis, chloride 82, HCO_3^- 38, K 2.8, creatinine 1.8, urine chloride low.

Teaching explanation: This is chloride-responsive metabolic alkalosis with hypokalemia and prerenal AKI from volume depletion. The low urine chloride links alkalosis to extrarenal chloride loss. Treat with isotonic saline and potassium chloride while monitoring renal recovery.

Case 4 - Hematuria after URI

Stem: A young adult develops gross hematuria two days after a sore throat. BP is mildly elevated, creatinine is slightly high, UA shows RBC casts and protein 1+.

Teaching explanation: RBC casts localize bleeding to glomeruli. The timing is synpharyngitic, favoring IgA nephropathy rather than classic post-streptococcal GN. Quantify proteinuria, check kidney function, assess risk, and involve nephrology if persistent proteinuria, reduced GFR, or severe presentation.

Case 5 - Pulmonary-renal syndrome

Stem: A patient has hemoptysis, anemia, creatinine rising over one week, dysmorphic RBCs/RBC casts, and new oxygen requirement.

Teaching explanation: This is rapidly progressive glomerulonephritis with possible pulmonary hemorrhage. Anti-GBM disease and ANCA vasculitis are high on the differential. This is not a wait-and-see hematuria; urgent serologies, nephrology, biopsy planning, and empiric therapy may be needed depending on severity.

Case 6 - Stone plus fever

Stem: A patient has colicky flank pain, fever, pyuria, creatinine rise, and CT shows an obstructing ureteral stone with hydronephrosis.

Teaching explanation: This is an infected obstructed collecting system. Antibiotics alone are insufficient because source control is required. Urgent decompression with ureteral stent or nephrostomy is the key management anchor.

Case 7 - Heavy protein and thrombosis

Stem: A patient presents with leg edema, albumin 2.0, urine protein/creatinine ratio nephrotic range, hyperlipidemia, and renal vein thrombosis.

Teaching explanation: This is nephrotic syndrome with hypercoagulability from urinary loss of anticoagulant proteins and hepatic lipoprotein synthesis. Membranous nephropathy is a classic adult association with renal vein thrombosis, but diabetic kidney disease, FSGS, amyloid, lupus, and malignancy-related disease remain in the differential.

Case 8 - Type 4 RTA pattern

Stem: A diabetic patient on lisinopril and spironolactone has K 6.0, HCO_3^- 18, normal anion gap, mild CKD, and no diarrhea.

Teaching explanation: Hyperkalemic non-anion-gap metabolic acidosis suggests impaired aldosterone effect: type 4 RTA. Treat emergent potassium if ECG changes, stop contributors, consider loop/thiazide diuresis, bicarbonate if acidotic, and address diabetic kidney disease/RAAS medication balance.

Case 9 - Contrast after dehydration

Stem: After CT contrast during a diarrheal illness, a patient develops creatinine rise over 48 hours with granular casts. FeNa is elevated.

Teaching explanation: This is likely ATN in a susceptible patient, but the key lesson is mixed risk: hypovolemia, contrast exposure, and possible nephrotoxin synergy. Management is supportive, with avoidance of further insults and monitoring for AEIOU complications.

Case 10 - Low complement nephritic disease

Stem: A patient has edema, hypertension, RBC casts, low C3, normal C4, and recent skin infection.

Teaching explanation: Low C3 with nephritic syndrome after infection supports postinfectious GN, though C3 glomerulopathy/MPGN remain considerations if complement does not normalize or disease persists. Treat infection/supportive care and monitor kidney function, blood pressure, and complement recovery.

Case 11 - CKD mineral bone disease

Stem: A patient with eGFR 22 has phosphate 6.2, calcium low-normal, PTH high, bone pain, and pruritus.

Teaching explanation: Reduced phosphate excretion and low calcitriol drive secondary hyperparathyroidism and renal osteodystrophy. Management links include phosphate restriction/binders, vitamin D-related therapy depending on context, and nephrology-directed CKD-MBD monitoring.

Case 12 - Polyuria after obstruction relief

Stem: After Foley placement for urinary retention, a patient produces 5 liters of urine in 24 hours and becomes lightheaded with falling potassium and magnesium.

Teaching explanation: Post-obstructive diuresis can cause volume depletion and electrolyte losses after decompression. Monitor urine output, daily weights, blood pressure, creatinine, sodium, potassium, magnesium, and replace fluid/electrolytes thoughtfully rather than assuming high urine output means full recovery.

6. Practice questions

1. A patient with AKI has urine Na 8 mEq/L, FeNa 0.4%, Uosm 620 mOsm/kg, bland urine sediment, orthostasis, and recent poor intake. What is the best classification?

- A. Acute interstitial nephritis
- B. Prerenal azotemia
- C. Postinfectious GN
- D. Nephrogenic DI

Answer: B. The urine is concentrated with avid sodium retention and bland sediment, supporting prerenal physiology.

2. A patient with AKI after aminoglycoside exposure has muddy brown casts and FeNa 3%. What is the most likely mechanism?

- A. Tubular epithelial injury
- B. Isolated low solute intake
- C. Antibody-mediated podocyte injury
- D. Pure water loss

Answer: A. Muddy brown casts and impaired sodium/water reabsorption point to ATN.

3. Serum Na is 118 with seizure and low serum osmolality. What is the immediate treatment priority?

- A. Rapid correction to normal sodium
- B. Hypertonic saline to improve severe symptoms while avoiding overcorrection
- C. D5W infusion
- D. Oral salt only

Answer: B. Severe symptomatic hypotonic hyponatremia requires hypertonic saline with frequent monitoring; full normalization is not the acute goal.

4. A patient has pH 7.25, HCO3- 12. Expected PaCO2 by Winter formula is closest to:

- A. 16-20 mm Hg
- B. 26-30 mm Hg
- C. 40-44 mm Hg
- D. 55-60 mm Hg

Answer: B. 1.5 x 12 + 8 = 26 +/- 2, so approximately 24-28; option B is closest.

5. Hematuria and RBC casts plus hemoptysis most strongly suggest which diagnostic frame?

- A. Uncomplicated cystitis
- B. Pulmonary-renal rapidly progressive GN syndrome
- C. Pure nephrotic syndrome
- D. Primary polydipsia

Answer: B. RBC casts plus lung hemorrhage points to anti-GBM disease or ANCA vasculitis until proven otherwise.

6. Nephrotic syndrome causes edema primarily through:

- A. ADH suppression and water loss
- B. Loss of albumin and renal sodium retention
- C. High serum oncotic pressure
- D. Respiratory alkalosis

Answer: B. Heavy proteinuria lowers oncotic pressure and activates sodium-retaining pathways.

7. Fever, flank pain, pyuria, hydronephrosis, and an obstructing ureteral stone require:

- A. Oral hydration only
- B. Antibiotics plus urgent decompression
- C. Immediate kidney biopsy
- D. Fluid restriction

Answer: B. Infected obstruction is a source-control emergency.

8. Hyperkalemia, non-anion-gap metabolic acidosis, diabetes, and RAAS blockade suggest:

- A. Type 4 RTA
- B. Central diabetes insipidus
- C. Vomiting-induced alkalosis
- D. Minimal change disease

Answer: A. Impaired aldosterone effect causes hyperkalemic non-gap acidosis.

9. In CKD, the CGA framework means:

- A. Creatinine, glucose, albumin
- B. Cause, GFR category, albuminuria category
- C. Calcium, glomeruli, acid-base
- D. Cystatin, GFR, age

Answer: B. CKD is classified by cause, GFR category, and albuminuria category.

10. A patient with severe kidney failure has pericarditis. Which dialysis indication applies?

- A. Acidosis only
- B. Uremia
- C. Mild asymptomatic creatinine elevation
- D. Low urine specific gravity

Answer: B. Uremic pericarditis is an urgent dialysis indication.

7. Common pitfalls and corrections

Pitfall	Why it is dangerous	Correction
Treating creatinine instead of the patient.	Creatinine lags behind GFR change and does not reveal arrhythmia, pulmonary edema, obstruction, or immune emergency.	Trend urine output, K, acid-base, volume status, sediment, and symptoms.
Assuming edema means intravascular overload.	Heart failure, cirrhosis, and nephrotic syndrome can trigger avid renal sodium/water retention from low effective arterial volume.	Separate total-body sodium/water from effective arterial blood volume.
Ignoring urine microscopy.	A "renal failure" label misses RBC casts, WBC casts, crystals, or ATN casts that change management.	Look at sediment early when AKI is unexplained.
Overusing FeNa.	Diuretics, CKD, contrast, GN, sepsis, and mixed disease can make FeNa misleading.	Use FeNa as one clue; consider FeUrea and the whole clinical picture.
Correcting chronic hyponatremia too fast.	Risk of osmotic demyelination, especially in malnutrition, alcoholism, liver disease, hypokalemia.	Use conservative correction goals and frequent sodium checks; relower if overcorrecting.
Missing obstruction in non-anuric patients.	Partial obstruction or unilateral obstruction in a solitary kidney can still produce urine early.	Use bladder scan/Foley and ultrasound when AKI is not clearly explained.
Calling all proteinuria diabetic kidney disease.	Nondiabetic glomerular disease may be treatable and can coexist with diabetes.	Question rapid onset, active sediment, short diabetes duration, absent retinopathy, systemic signs, low complement.
Waiting for serologies in pulmonary-renal syndrome.	RPGN can progress rapidly and pulmonary hemorrhage can be fatal.	Urgent nephrology, serologies, biopsy planning, and timely immunologic treatment decisions.

8. Graph-RAG extraction map

For graph retrieval, the renal section should not store isolated definitions only. It should store relationships: mechanism-to-lab, lab-to-syndrome, syndrome-to-emergency, disease-to-complication, and treatment-to-risk. The highest value renal graph edges are often conditional: a urine sodium of 10 has different meaning in heart failure, vomiting, diuretic use, and ATN recovery.

Core node classes

Node class	Examples	Key properties to store
Syndrome	Hyponatremia, hypernatremia, metabolic acidosis, AKI, CKD, nephritic syndrome, nephrotic syndrome, renal colic.	Definition thresholds, danger signs, initial tests, common mechanisms, management priorities.
Mechanism	Low effective arterial volume, ADH excess, aldosterone deficiency, tubular necrosis, immune complex deposition, podocyte injury, urinary obstruction.	Input triggers, expected labs, expected urine findings, reversibility, complications.
Disease	SIADH, DKA, ATN, AIN, IgA nephropathy, lupus nephritis, ANCA vasculitis, membranous nephropathy, ADPKD, nephrolithiasis.	Illness script, demographics, triggers, labs, urine, imaging, biopsy, treatment.
Finding	RBC casts, muddy brown casts, low C3, urine Na <20, FeNa <1%, high anion gap, osmolar gap, hydronephrosis.	What it supports, what it does not exclude, confounders.
Treatment	Hypertonic saline, IV calcium, insulin/dextrose, isotonic fluids, loop diuretics, RAAS blockade, immunosuppression, dialysis, decompression.	Indication, contraindications, monitoring, adverse effects, linked emergencies.
Complication	Osmotic demyelination, arrhythmia, cerebral edema, pulmonary edema, uremic pericarditis, renal osteodystrophy, thrombosis, infection.	Risk factors, prevention, early signs, management.

High-value edges

Edge	Relationship statement
Low effective arterial volume -> ADH secretion	Explains hyponatremia in CHF/cirrhosis/nephrotic syndrome despite edema.
ADH activity -> high urine osmolality	Urine cannot dilute appropriately when ADH is active.
Hypokalemia -> metabolic alkalosis maintenance	Low K promotes renal H+ secretion and bicarbonate retention.
Acidosis -> extracellular K shift	Especially mineral acidosis; worsens hyperkalemia risk in AKI/CKD.
ATN -> muddy brown granular casts	Tubular cell injury creates pigmented granular sediment.
Glomerular capillary inflammation -> RBC casts	RBC casts localize hematuria to the nephron.
Nephrotic syndrome -> thrombosis	Urinary loss of anticoagulant proteins plus hepatic synthesis changes increases clot risk.
Low complement + nephritic syndrome -> immune complex/complement disease	Supports postinfectious GN, lupus nephritis, MPGN/C3, cryoglobulinemia depending on C3/C4 pattern.
Obstruction + infection -> urgent decompression	Antibiotics require source control when infected urine is trapped under pressure.
Kidney failure + refractory overload -> dialysis	AEIOU links clinical emergency to renal replacement therapy rather than creatinine alone.

Suggested retrieval anchors

Anchor term	Synonyms / linked phrases
Prerenal azotemia	low effective arterial volume, FeNa <1%, high BUN/Cr, concentrated urine, hyaline casts

Anchor term	Synonyms / linked phrases
ATN	muddy brown casts, granular casts, ischemic tubular injury, nephrotoxic AKI, FeNa high
RPGN	crescentic GN, pulmonary-renal syndrome, ANCA, anti-GBM, RBC casts, rapidly rising creatinine
SIADH	euvolemic hyponatremia, high urine osmolality, high urine sodium, ADH excess
Type 4 RTA	hyperkalemic non-gap acidosis, hypoaldosteronism, diabetic kidney disease, ACE inhibitor, spironolactone
Nephrotic syndrome	proteinuria >3.5 g/day, edema, hypoalbuminemia, hyperlipidemia, lipiduria
CKD-MBD	hyperphosphatemia, secondary hyperparathyroidism, low calcitriol, renal osteodystrophy
Obstructive uropathy	hydronephrosis, urinary retention, BPH, Foley, bladder scan, post-obstructive diuresis

9. Final renal synthesis checklist

Before leaving any renal case, answer these questions in order. This checklist catches most exam and clinical misses.

Question	Why it matters
1. Is the patient unstable from K, sodium, pH, pulmonary edema, toxin, uremia, infection, or obstruction?	Determines whether treatment starts before the workup is complete.
2. Is the sodium problem a tonicity problem, and what is the correction risk?	Prevents missing pseudo/hypertonic hyponatremia and prevents osmotic injury.
3. Is potassium dangerous on ECG or likely to worsen?	Avoids fatal arrhythmia while causes are evaluated.
4. Does acid-base compensation fit?	Detects mixed disorders such as sepsis plus renal failure, COPD plus vomiting, or salicylate toxicity.
5. Is kidney dysfunction acute, chronic, or acute-on-chronic?	Changes prognosis, imaging interpretation, and search for reversible triggers.
6. Is the urine sediment bland, tubular, glomerular, inflammatory, or crystalline?	Localizes disease and prioritizes nephrology/biopsy/imaging.
7. Has obstruction been excluded when appropriate?	Postrenal AKI is reversible but dangerous when missed.
8. Are medications causing injury or requiring renal dose adjustment?	Prevents ongoing harm.
9. Is this nephritic, nephrotic, or mixed glomerular disease?	Directs complement/serology/biopsy logic.
10. Are AEIOU dialysis indications present?	Creatinine alone is not enough; symptoms and refractory derangements drive urgency.

Chapter close

Renal medicine rewards mechanism-based reasoning because the same few physiologic levers explain many presentations. ADH explains water retention and urine concentration. RAAS and aldosterone explain sodium retention, potassium excretion, and effective arterial blood volume. The glomerular barrier explains protein, blood, edema, thrombosis, and immune clues. Tubules explain concentration, acid-base, potassium, magnesium, phosphate, and many drug toxicities. Drainage explains why a mechanical problem can look like kidney failure. A strong renal differential is therefore not a long list; it is a map from patient clues to nephron function.

Selected references and accuracy checks

- KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. Kidney International. 2024.
- KDIGO Clinical Practice Guideline for Acute Kidney Injury. Kidney International Supplements. 2012.
- KDIGO 2021 Clinical Practice Guideline for the Management of Glomerular Diseases. Kidney International. 2021.
- KDIGO 2024 Clinical Practice Guideline for the Management of Lupus Nephritis. Kidney International. 2024.
- KDIGO 2024 Clinical Practice Guideline for the Management of ANCA-Associated Vasculitis. Kidney International. 2024.
- European Association of Urology Guidelines on Urolithiasis. 2024 update.
- European Society of Endocrinology clinical practice guidance on management of hyponatraemia; emergency hyponatraemia safety principles cross-checked for correction limits and monitoring.

Educational note: This resource is for medical learning and should be reconciled with local protocols, specialist guidance, and patient-specific factors in clinical care.